

Academic Reflection Report: Introduction to Natural Language Processing Concepts

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Abstract—This report is a formal reflection about the module: Introduction to natural language processing concepts in Microsoft learn. It explains how to move from structured data processing to modern unstructured text analytics. In the hands-on text analytics exercises of Unit 5, the reflection will discuss tokenization, TF-IDF, embeddings, and context vectors.

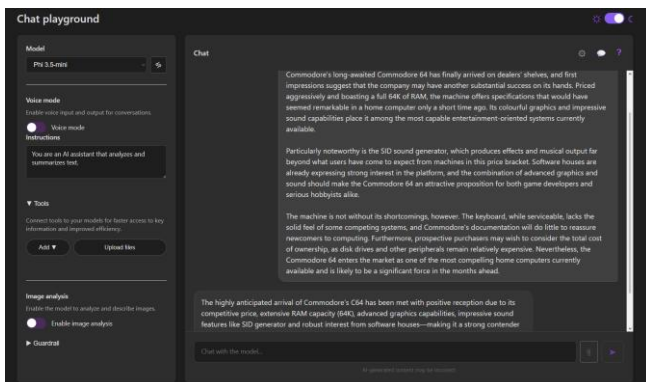
Keywords—NLP, token, analytic, text, machine learning

I. INTRODUCTION

This is a major paradigm shift from structured data analysis, with unstructured text data being processed. Before finishing the Microsoft Learn natural language processing (NLP) module, the concepts of data processing were largely centered around the structured database, traditional query logic and tabular predictive modeling. They work well with numerical and categorical data but are not readily applicable to raw human language. The module called “Introduction to natural language processing concepts” gave a detailed introduction to the field of processing unstructured data. The module covered the basic text cleaning, up to the advanced semantic representation, which is the pipeline of NLP. This report summarizes the hands-on experience with the module, and discusses the trade-offs between various text representation strategies from a technical perspective.

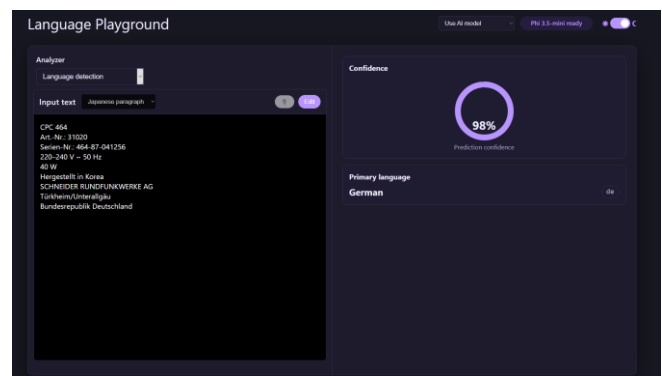
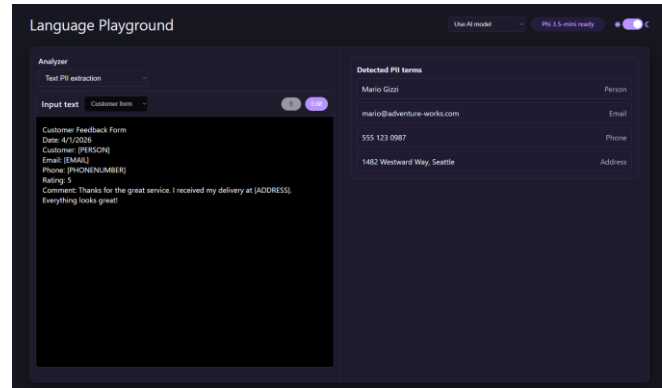
II. EXPERIENTIAL & PRACTICAL INSIGHTS

Unit 5 of the module featured hands-on text analytics exercises that transitioned theoretical concepts into practical implementation. In this exercise, I used Microsoft’s Chat Playground and Language Playground app. Using the Chat Playground app, I used the generative AI to generate a summary of a sample article. First, I changed the instruction for the AI for it to analyze and summarize texts. Then, I proceeded to copy and paste the sample article from the exercise to generate the summarize version using the AI.



The next part of the exercise uses the Language Playground app to identify the language (e.g. English, Japanese, German) using the Language Detector and using

the Text PII extraction to provide the terms that can be link to a Personally Identifiable Information.



III. CORE LEARNING OUTCOMES

A. Tokenization Trade-offs

The first step in the NLP pipeline is the tokenization step, which is important, but comes with its own inherent trade-offs. The word level tokenization is easy to understand, but has poor performance with out-of-vocabulary (OOV) words and needs a huge vocabulary. On the other hand, OOV issues are completely eliminated with character level tokenization, but sequences are too long, losing semantic information. The module presented how the utilization of subword tokenization (such as Byte-Pair Encoding) could offer a powerful middle ground between vocabulary size and semantic retention, which is suitable for a morphologically rich language.

B. TF-IDF vs. Dense Embeddings

The module was clearly separating the evolutionary step between Term Frequency-Inverse Document Frequency (TF-IDF) and word embeddings. Though TF-IDF vectors are sparse and high dimensional, they reflect the statistical significance of a word in a document, but do not reflect relationships. With dense embeddings, the words are represented as a lower-dimensional continuous vector space

in which the geometric distance is proportional to the degree of semantic similarity. For me, one of the key technical breakthroughs was realizing that, if a model could learn that two conceptually similar words correspond to similar positions in an embedding, they could also learn that two conceptually different words correspond to orthogonal dimensions in an embedding; TF-IDF did the latter while the embedding did not.

C. Context Vectors

Static word representations capture the dynamic nature of the word, however, they do not model polysemy (words with multiple meanings). In the introduction to context vectors, the idea was explained of how a word's numerical representation can change depending on the tokens around it, using the self-attention mechanism. Context vectors were important to understand because they were essential for comprehending sequences as a whole as opposed to as single tokens in modern architecture.

IV. COURSE APPLICATION & IMPACT

The use of these NLP concepts will directly contribute to my future academic accomplishments, especially in the deep learning course. Access to cloud-based text analytics services can enable intelligent enhancements to existing IT applications, such as automating support ticket routing or analyzing customer feedback. The dense embeddings and text classification enable the automatic categorization of incoming unstructured data according to the semantic meaning of the text, not just by the presence of simple keywords, which can greatly improve the efficiency of the operation. In addition, the module highlighted the need to perform careful validation when these NLP derived features were combined with powerful classification models. The evaluation modules for these models need to be programmed explicitly, to be in line to the correct evaluation standards for future labs. The validation logic should perform calculations directly (e.g. explicitly use accuracy score for test sets), without trying to use abstract pipeline functions that obfuscate the sequence of evaluations. It guarantees control of text embedding and natural language processing features on the resultant predictive performance.

V. CONCLUSION & FUTURE WORK

The Microsoft Learn module gave a firm foundation for understanding the basics of text analytics, and how to get from statistical term counts to semantic vector representation. The practical problems of text normalization introduced in Unit 5 were reinforced through the hands-on activities and the cloud-based NLP APIs were impressive. Going forward, future study areas will be centered around tuning up Large Language Models (LLMs) and delving into the depths of Transformer architectures. This knowledge base will be expanded to develop more context-aware, intelligent applications capable of bridging the human to automated system gap.